

The function and the ways of defining it

The same function written five ways — formula, table, graph, words and a machine — and why a question sometimes wants one rather than another.

LESSON 9–10

2 × 40 MIN

ALGEBRA 10, §1.2

P1 · 2.1–2.2

LESSON CLOCK

6'

12'

8'

12'

7'

5'

Recall — exactly one output	6 min
Five ways of writing one function	12 min
Function notation	8 min
The vertical line test	12 min
Practice	7 min
Homework	5 min

LEARNING OBJECTIVES

- State the definition of a function in terms of domain, rule and image.
- Move between formula, table, graph and verbal description of the same function.
- Use function notation $f(x)$ and evaluate f at a number and at an expression.
- Apply the vertical line test to a graph.

TEXTBOOK

National	pp. 28–34
Cambridge	Pure Mathematics 1, pp. 26–33
Workbook	P1 Exercise 2A

01

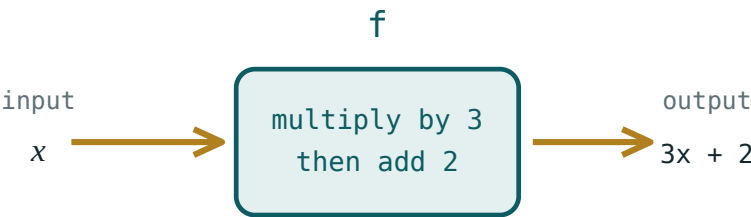
Explanation

What has to be given

A FUNCTION NEEDS THREE THINGS

A **domain** — the set of allowed inputs. A **rule** — what to do to an input. And the guarantee that the rule gives **exactly one** output for each input.

Change the domain and you change the function, even if the rule is unchanged. $f(x) = x^2$ on $\mathbb{R}$ is many-to-one; the same rule on $x \geq 0$ is one-to-one. They are different functions.



*The machine picture: one input in, the rule applied,
exactly one output out.*

Five ways to write the same function

The doubling-and-adding-one function, five ways:

WAY	HOW IT LOOKS	BEST WHEN
formula	$f(x) = 2x + 1$	you must compute or manipulate
table	$0 \mapsto 1, 1 \mapsto 3, 2 \mapsto 5$	the domain is small or the data is measured
graph	a straight line, gradient 2	you want shape, roots, behaviour
words	“double the number, then add one”	explaining, or modelling a real situation
ordered pairs	$\{(0,1), (1,3), (2,5)\}$	the definition must be checked exactly

A TABLE IS NOT ALWAYS ENOUGH

Three points do not determine a function. $f(x) = 2x + 1$ and a cubic through the same three points agree on the table and disagree everywhere else. A table defines a function only when the domain is exactly the listed inputs.

Function notation

$f(3)$ means “the output when the input is 3”. It is not multiplication. The letter inside is a placeholder, so it can be replaced by anything — a number, another letter, a whole expression.

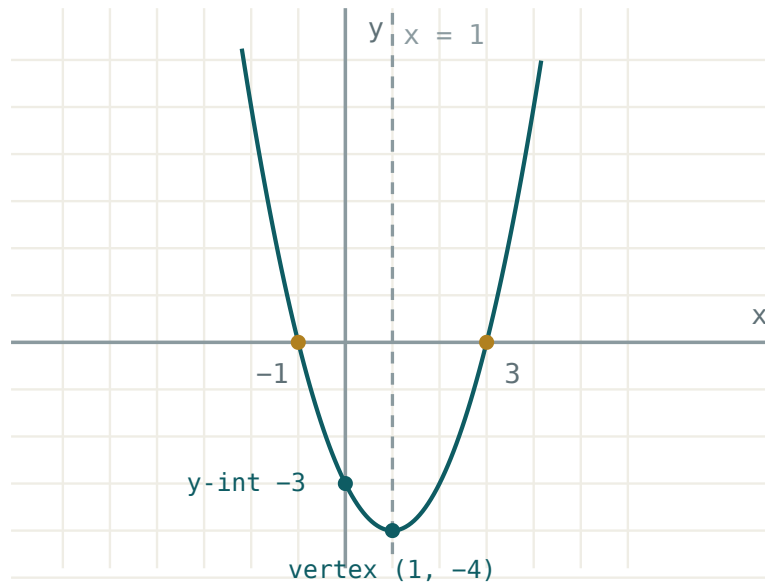
$$f(x) = x^2 - 3x \Rightarrow f(2) = -2, f(a) = a^2 - 3a, f(x + 1) = (x + 1)^2 - 3(x + 1)$$

SUBSTITUTE THE WHOLE THING

$f(x + 1)$ is **not** $f(x) + 1$. Put brackets round the substituted expression every time, then expand. Most lost marks in this topic live in that one missing bracket.

The vertical line test

A graph shows a function exactly when no vertical line meets it more than once — that is the picture of “exactly one output per input”.



Every vertical line meets this parabola once. It is a function — many-to-one, but a function.

A circle fails: the line $x = 0$ meets $x^2 + y^2 = 25$ at $(0, 5)$ and $(0, -5)$. So a circle is a relation, not a function of x .

02 Worked examples

EXAMPLE 1 For $f(x) = x^2 - 4x + 1$, find $f(0)$, $f(-2)$ and $f(a + 1)$.

① $f(0) = 0 - 0 + 1 = 1$

② $f(-2) = 4 + 8 + 1 = 13$
Watch $-4 \cdot (-2) = +8$.

③ $f(a + 1) = (a + 1)^2 - 4(a + 1) + 1$
Brackets first.

④ $= a^2 + 2a + 1 - 4a - 4 + 1 = a^2 - 2a - 2$

ANSWER

$1, 13, a^2 - 2a - 2$

EXAMPLE 2 If $g(x) = 3x - 5$ and $g(k) = 7$, find k .

① $3k - 5 = 7$
Read the notation as an equation.

② $3k = 12$

③ $k = 4$

ANSWER

$k = 4$

EXAMPLE**3**

A taxi charges 8000 so‘m plus 2500 so‘m a kilometre. Write the cost as a function and find the cost of 12 km.

1

Words to formula.

$$C(d) = 8000 + 2500d$$

2

State the domain.

$$d \geq 0$$

3

$$C(12) = 8000 + 30000 = 38000$$

ANSWER

$$C(d) = 8000 + 2500d; 38\,000 \text{ so‘m}$$

03 Interactive model

Change the rule and watch the table, the graph and the machine all update together.

Reading function notation

If $f(x) = 5 - 2x$ then $f(4)$ is:

If $f(x) = x^2$ then $f(x + 2)$ is:

A vertical line cutting a graph twice means:

If $h(x) = \underline{1} x - 3$ then $h(3)$ is:

0

1

undefined

3

$f(x) = 4$ for all x has a graph that is:

a vertical line

a horizontal line

a parabola

not a graph

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Function	Funksiya	Функция
Function notation	Funksional belgilash	Функциональное обозначение
Independent variable	Erkli o'zgaruvchi	Независимая переменная
Dependent variable	Erksiz o'zgaruvchi	Зависимая переменная
Rule	Qoida	Правило
Table of values	Qiyimatlar jadvali	Таблица значений
Graph of a function	Funksiya grafigi	График функции
Vertical line test	Vertikal chiziq sinovi	Тест вертикальной прямой
Piecewise definition	Bo'lakli berilish	Кусочное задание

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$f(x) = 2x + 3$, so $f(-1)$ is:

The independent variable in $y = f(x)$ is:

Which cannot be the graph of a function?

If $f(x) = x^2$, then $f(a) = f(b)$ means:

$$a = b$$

$$a = \pm b$$

$$a = -b$$

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $f(x) = 3x + 1$; find $f(2)$

ANSWER 7

2. $f(x) = 3x + 1$; find $f(0)$

ANSWER 1

3. $f(x) = x^2 - 1$; find $f(-3)$

ANSWER 8

4. $g(x) = 10 - x$; find $g(10)$

ANSWER 0

5. Is a circle the graph of a function?

ANSWER no

6. $f(x) = 7$; find $f(100)$

ANSWER 7

7. Write “treble it, then subtract 2” as a formula

ANSWER $f(x) = 3x - 2$

Medium · the standard question

7 PROBLEMS

1. $f(x) = x^2 - 2x$; find $f(a + 1)$

ANSWER $a^2 - 1$

2. $f(x) = 4x - 9$; solve $f(x) = 0$

ANSWER $x = 2.25$

3. $f(x) = \frac{x}{x + 2}$; find $f(2)$

ANSWER 0.5

4. $f(x) = \frac{x}{x + 2}$; which input is forbidden?

ANSWER $x = -2$

5. $f(x) = x^3$; solve $f(x) = 8$

ANSWER $x = 2$

6. A phone plan costs 30 000 plus 200 a minute. Write $C(t)$

ANSWER $C(t) = 30000 + 200t$

7. $f(x) = 2x$; show $f(a + b) = f(a) + f(b)$

ANSWER $2(a + b) = 2a + 2b$ ✓

Hard · stretch and reason

7 PROBLEMS

1. $f(x) = x^2 + 1$; simplify $\frac{f(x+h) - f(x)}{h}$

ANSWER $2x + h$

2. $f(x) = \frac{1}{x}$; simplify $f(a) - f(b)$

ANSWER $\frac{b-a}{ab}$

3. $f(x) = 2x + 3$; find x with $f(x) = f(2x)$

ANSWER $x = 0$

4. $f(x) = x^2$; find all a with $f(a) = f(a+2)$

ANSWER $a = -1$

5. Give a rule whose graph fails the vertical line test

ANSWER $x = y^2$

6. $f(x) = ax + b$, $f(1) = 5$, $f(3) = 11$. Find a , b

ANSWER $a = 3$, $b = 2$

7. Explain why $f(x+1) \neq f(x) + 1$ in general

ANSWER Only additive rules satisfy it — e.g. $f(x) = x^2$ fails

67 Homework

Homework — 5 tasks

Task 5 needs the vertical line test drawn, not just named.

- $f(x) = x^2 - 5x + 6$. Find $f(0)$, $f(2)$, $f(-1)$.
- $f(x) = 3 - 4x$. Solve $f(x) = 15$.
- $f(x) = x^2 + x$. Simplify $f(x+1) - f(x)$.
- Write the same function four ways: formula, table on $\{0,1,2,3\}$, words and ordered pairs.

5. Sketch a curve that is a function and one that is not, and mark the vertical line that decides each.